

PULSE AI: Preliminary Requirements Document (PRD)

Baseline Functional, Non-Functional, Biometric & Architecture Requirements

Document Identifier: PRD-PULSE-2026-V1	Lifecycle Phase: Preliminary Design & Verification
Target Platform: Cross-Platform Web & Mobile PWA	System Core: Next.js 14 + MediaPipe + FastAPI
Privacy Classification: Zero-Cloud Video Transmission	Operational Status: Baseline Requirements Validated

1. Product Vision & Target Personas

Product Vision: PULSE AI democratizes professional personal training by combining edge computer vision form validation with conversational motivation and macro nutrition, delivering an instantaneous, private, and hyper-smooth workout companion without expensive wearable hardware.

- **Persona 1 (Home Workout Enthusiast):** Needs real-time form correction for bodyweight squats without latency or video privacy concerns.
- **Persona 2 (Fat Loss / Hypertrophy Athlete):** Needs precise macronutrient distributions (protein, carb, fat targets) and structured weekly groceries matching clinical BMI baselines.
- **Persona 3 (Busy Professional / Beginner):** Needs conversational guidance tailored to daily emotional states (e.g. overcoming fatigue or lack of motivation) and verified nearby gym recommendations.

2. Functional Requirements Matrix

Req ID	Module	Functional Requirement Description	Priority	Status
FR-01	AI Pose Coach	The system shall stream local HTML5 webcam input to an in-browser MediaPipe Pose detector, tracking Left Hip (#23), Left Knee (#25), and Left Ankle (#27) without backend streaming.	MUST	VERIFIED
FR-02	Biomechanics	The system shall calculate knee joint angle in real-time using 2D trigonometric vectors and register reps when angle transitions from DOWN (<95°) to UP (>160°).	MUST	VERIFIED
FR-03	Audio Feedback	The system shall trigger browser Web SpeechSynthesis vocal cues ('Good Rep!', 'Go Lower') on state transitions, with a user-facing mute/unmute control.	MUST	VERIFIED
FR-04	Virtual Buddy	The system shall provide a conversational chat interface communicating with FastAPI POST /api/chat, classifying user mood (PUMPED, FATIGUED, UNMOTIVATED, FOCUSED).	MUST	VERIFIED
FR-05	AI Dietician	The system shall ingest user biometrics (weight, height, goal, preference, calories) via POST /api/diet, returning BMI category, macro grams, and an interactive grocery checklist.	MUST	VERIFIED
FR-06	Gym Locator	The system shall query regional gym databases by city/area via POST /api/gym, providing verified star ratings, daily routines, and gamified challenges.	SHOULD	VERIFIED
FR-07	Mobile PWA	The system shall support Web App Manifest installation to home screens on iOS Safari and Android Chrome, rendering standalone fullscreen without browser chrome.	SHOULD	VERIFIED

3. Non-Functional Requirements (NFR)

NFR ID	Category	Requirement & Target Benchmark	Validation Method
NFR-01	Latency & Frame Rate	Client-side pose detection must maintain a continuous frame rate of ≥ 30 FPS with inference latency ≤ 33 ms per frame on consumer hardware.	Empirical benchmarking: achieved ~16.2 ms (60 FPS).
NFR-02	Zero-Knowledge Privacy	Raw webcam video streams must never be transmitted across networks or stored on external servers. All video parsing must occur exclusively in memory.	Architectural sandbox: MediaPipe WebAssembly execution.
NFR-03	API Performance	FastAPI backend endpoints (/api/chat, /api/diet, /api/gym) must return JSON payloads within ≤ 50 ms under concurrent local requests.	Measured round-trip: < 12 ms average.
NFR-04	Network Security	All public mobile endpoints must be encrypted using TLS 1.3 / HTTPS to satisfy mobile browser camera security standards.	Cloudflare Tunnel edge SSL termination.
NFR-05	Responsive Ergonomics	UI must gracefully adapt across desktop (sidebar layout) and mobile viewports (portrait-first bottom navigation) with dark-mode aesthetic.	Tailwind CSS fluid grid & glassmorphism.

4. Hardware & Software Operating Environment

- **Client Environment:** Modern Chromium / WebKit browser (Chrome 110+, Safari 16+, Edge 110+) with WebGL 2.0 and WebAssembly enabled. Integrated or USB 720p/1080p webcam.
- **Backend Runtime:** Python 3.10 to 3.14, FastAPI 0.141+, Starlette 1.7+, Uvicorn 0.53+, Pydantic 2.13+.

- **Frontend Runtime:** Node.js v18 to v24 LTS, Next.js 14.2+, React 18.3+, Tailwind CSS 3.4+, Framer Motion 11.11+.
- **Proxy Gateway:** Next.js Server-Side Route Handlers (/api/py/*) reverse-proxying local uvicorn sockets to prevent mixed-content blocks over public HTTPS tunnels.

5. Acceptance Criteria & Approval Sign-Off

Acceptance Criteria: 1. User can track 10 squats consecutively with accurate +1 increment upon achieving <95° depth. 2. Web Speech API emits audible cues for 'Good Rep' and 'Go Lower' when unmuted. 3. Diet calculator displays correct clinical BMI category, strategy, and grocery items. 4. Chatbot correctly parses emotional keywords and suggests actions. 5. Mobile users can install app to Home Screen via Cloudflare HTTPS URL.	Requirements Sign-Off: Product Manager: Approved Lead Engineer: Approved QA Lead: Passed All Criteria Date: September 25, 2026
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